Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 48

Concentration and Dilution of Urine

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0323. Predict the changes in body fluid volume and osmolality caused by a net NaCl or water loss/gain in the body.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0377. Predict how changes in body fluid volume and osmolality alter the rate of urine production and the osmotic composition of the urine.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0324. Identify the role of ADH release that allows either a dilute or concentrated urine to be produced and excreted.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0325. Describe the syndrome of inappropriate ADH release (SIADH) and contrast its mechanism of ADH release to normal stimuli.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0326. Explain the role of the ascending limb of the loop of Henle in producing a high renal interstitial fluid osmolality.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0378. Beginning with the loop of Henle, compare and contrast the tubular fluid and interstitial fluid osmolarity changes that allow either a dilute or concentrated urine to be produced and excreted.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0327. Explain the tubular section and cellular mechanism by which ADH increases permeability to water and urea.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0379. Describe the role of these changes on the ability of the kidney to produce either a dilute or concentrated urine.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0328. Distinguish between central and nephrogenic diabetes insipidus based on plasma ADH levels and the response to an injection of ADH.

Recommended Reading

Medical Physiology: Principles for Clinical Medicine with Companion Website Access, R. Rhoades & D. Bell 6th ed., 2023; Chapter 22

Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, D. Golan, E. J. Armstrong, & A. W. Armstrong, 4th ed., 2016; p. 474

Plasma and Urine Osmolarity

Plasma osmolarity is approximately 290 mOsm/L (300 mOsm/L for calculation purposes).

- By excreting a dilute or concentrated urine plasma osmolarity is maintained (osmoregulation)
- The excretion or retention of water plays a key role in osmoregulation and depends on antidiuretic hormone (ADH) levels.
- High ADH: Retention of water, concentrated urine
- Low ADH: Excretion of water, dilute urine

Osmolality – mOsm/kg

Urine osmolarity ranges from about 50 to 1200 mOsm/L

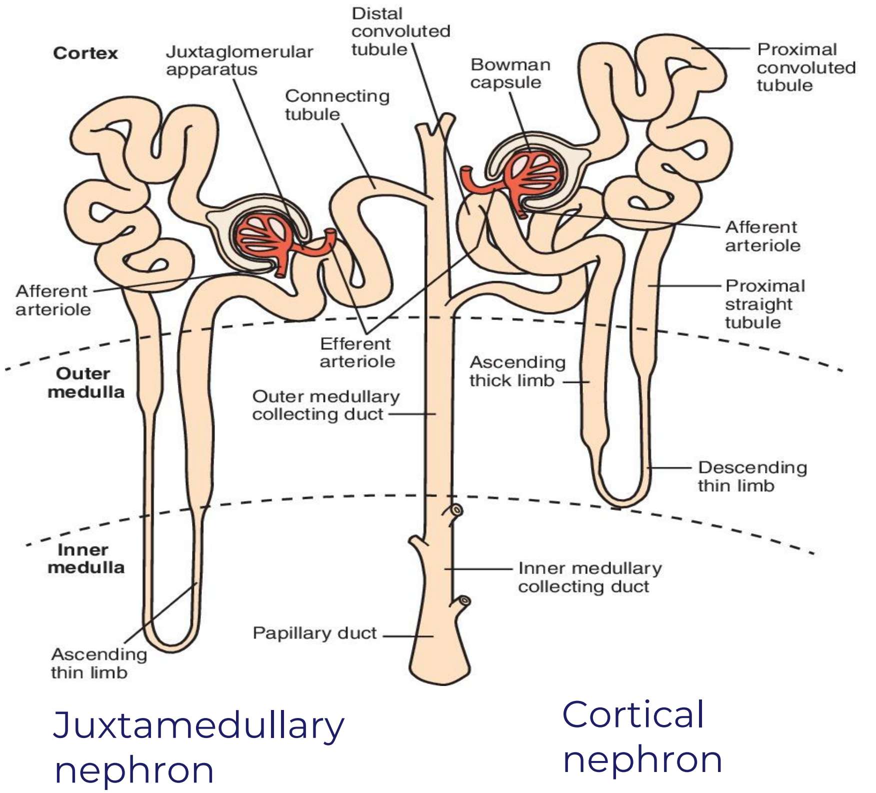
- Isosmotic urine – 300 mOsm/L
- Values more than 300 mOsm/L – hyperosmotic.
- Values less than 300 mOsm/L - hypoosmotic.

Concentration or Dilution of Urine by the Kidney

The kidneys must excrete approximately 600mOsm of solute wastes daily.

A minimum urine volume of 0.5 L/day is required to achieve this, assuming the kidneys can concentrate urine to a maximum of 1200 mOsm/L. (600 mOsm/day **/** 1200 mOsm/L = 0.5L/day).

This minimum urine volume is termed obligatory volume.



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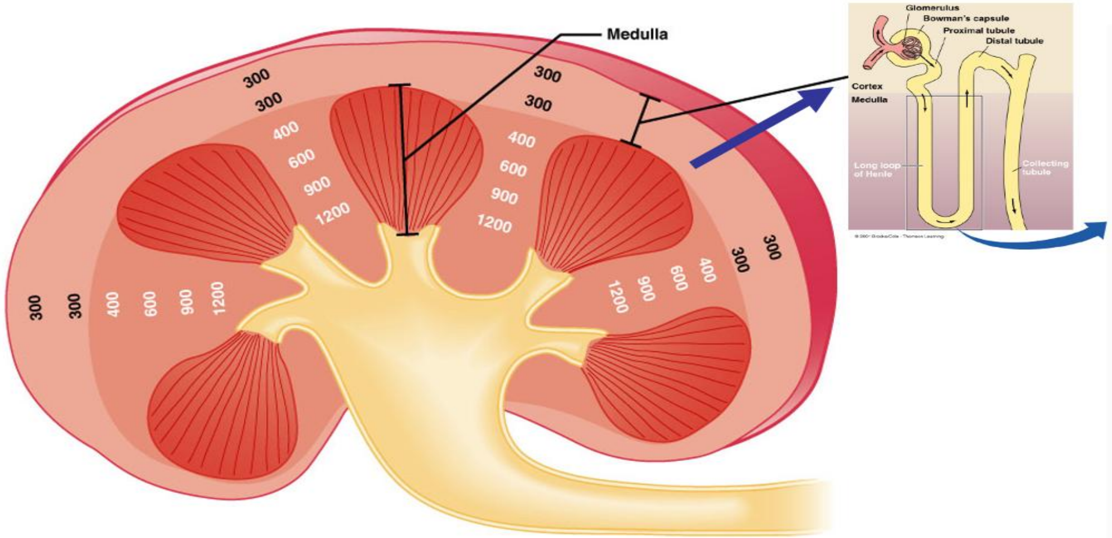
Generation of Corticomedullary Osmotic Gradient



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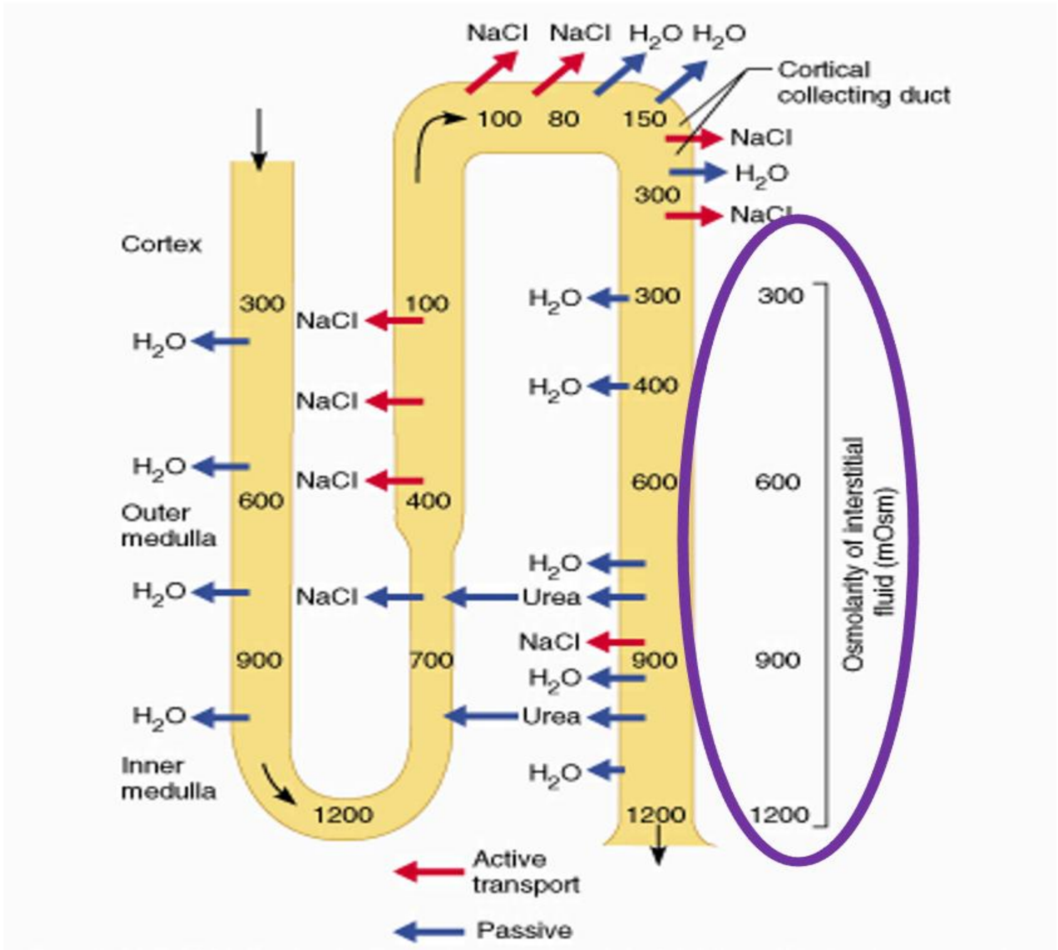
Processes that contribute to the corticomedullary osmotic gradient:

- Countercurrent multiplier
- Urea recycling
- Slow flow rates

This gradient is crucial for the concentrating and diluting ability of the kidney.



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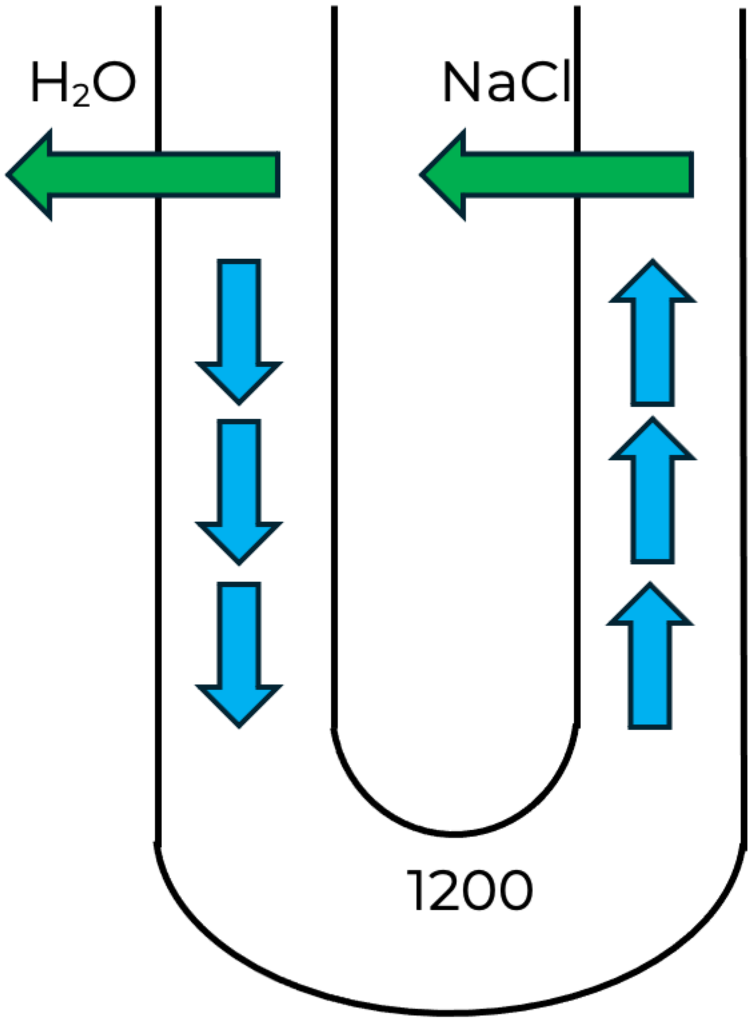
Countercurrent Multiplication

Recall features of the LOH:

- The descending limb of LOH is permeable to water and impermeable to NaCl
- The ascending limb of LOH is permeable to NaCl and impermeable to water.

The countercurrent multiplication depends on the **single effect** (movement of NaCl out of the thick ascending LOH and into the interstitium, as well as the subsequent movement of water out of the descending LOH into the interstitium) and **fluid flow** (new tubular fluid entering the descending LOH pushes the fluid with higher osmolality down the tube, resulting in the development of the osmotic gradient).



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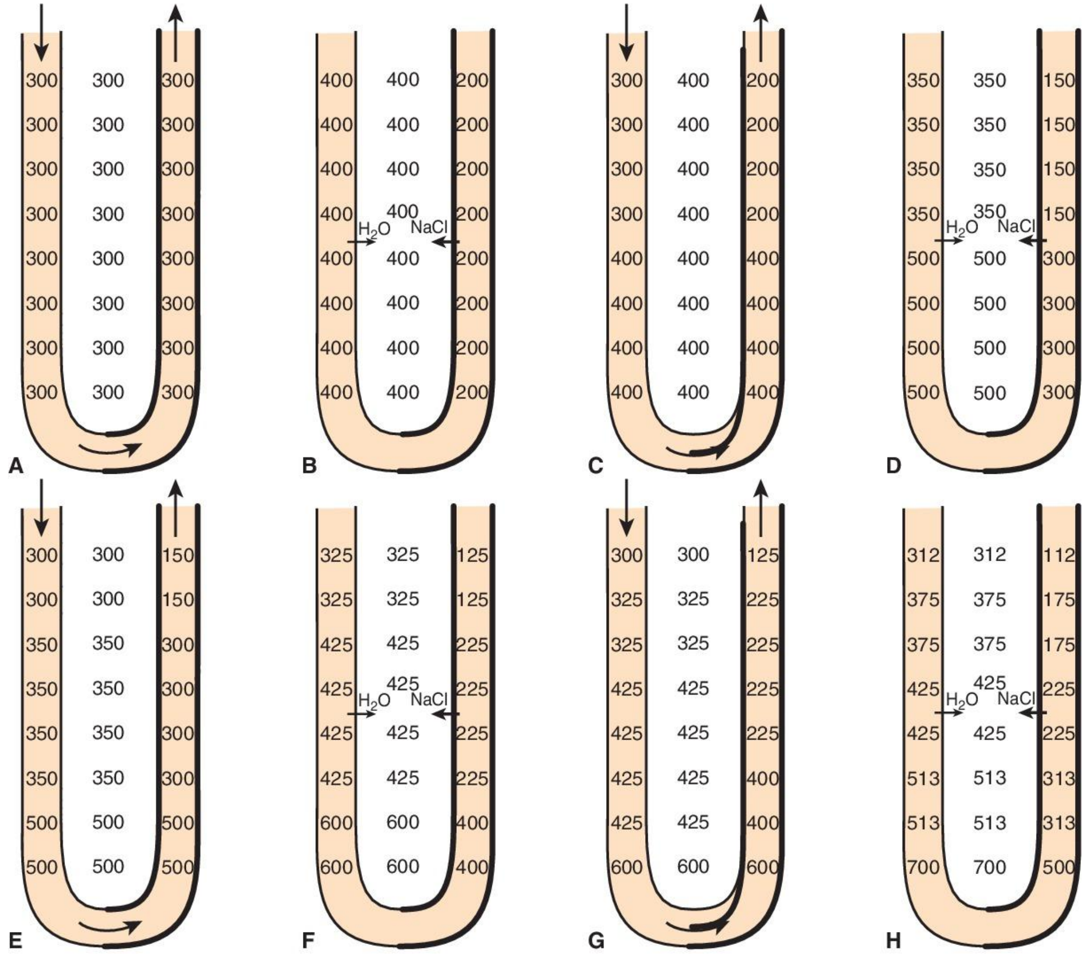
Countercurrent Multiplication



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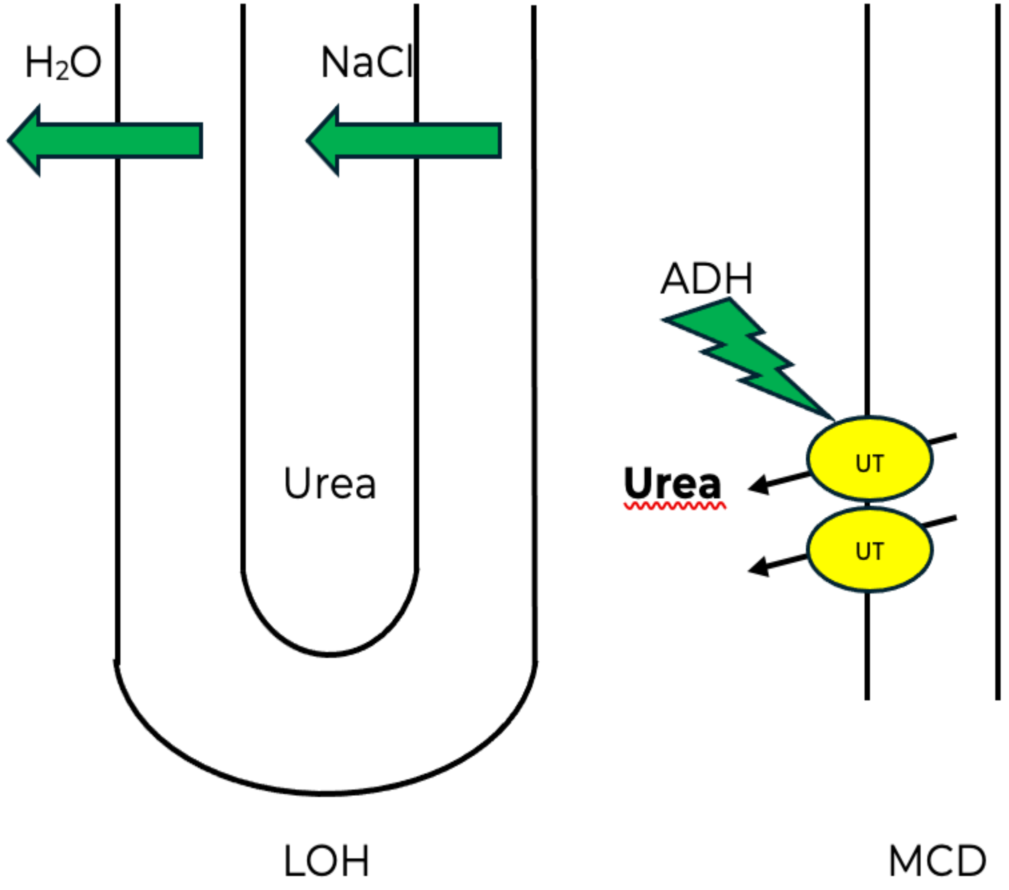
Urea Recycling

Urea plays a significant role in the osmolality of the inner medullary peritubular (interstitial) fluid.
It contributes to about 50% of the osmotic gradient in this region.

ADH upregulates urea transporters (UT), specifically UTA1 and UTA3, in the inner medullary collecting duct (MCD).

Urea diffuses into the interstitium, with some finding their way back into the LOH–**urea recycling**.



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Urea Concentration and Dietary Protein

Dietary protein influences urea concentration in the renal medulla.

Urea is a by-product of protein metabolism.

High protein in the diet increases the concentration of urea in the medullary interstitium, increasing the ability of the kidney to concentrate urine.

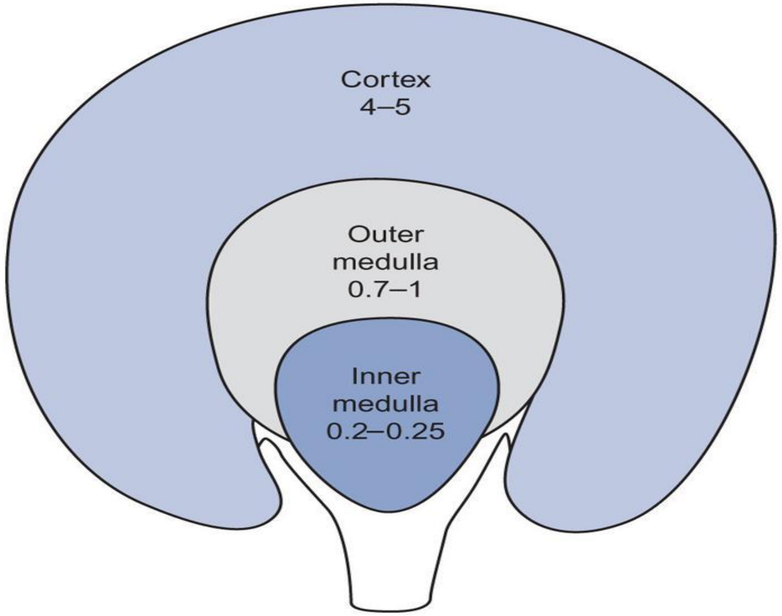
Conversely, low protein intake decreases the kidneys' ability to concentrate urine.

Newborns have decreased urine concentrating abilities but begin to concentrate urine like adults at about 12 months of age.

Preservation of Corticomedullary Osmotic Gradient

The **blood flow rates**, which decrease progressively from the cortex to the medulla, and the **vasa recta (countercurrent exchanger)** help preserve the generated corticomedullary osmotic gradient.

Blood flow rates (ml/min/g)



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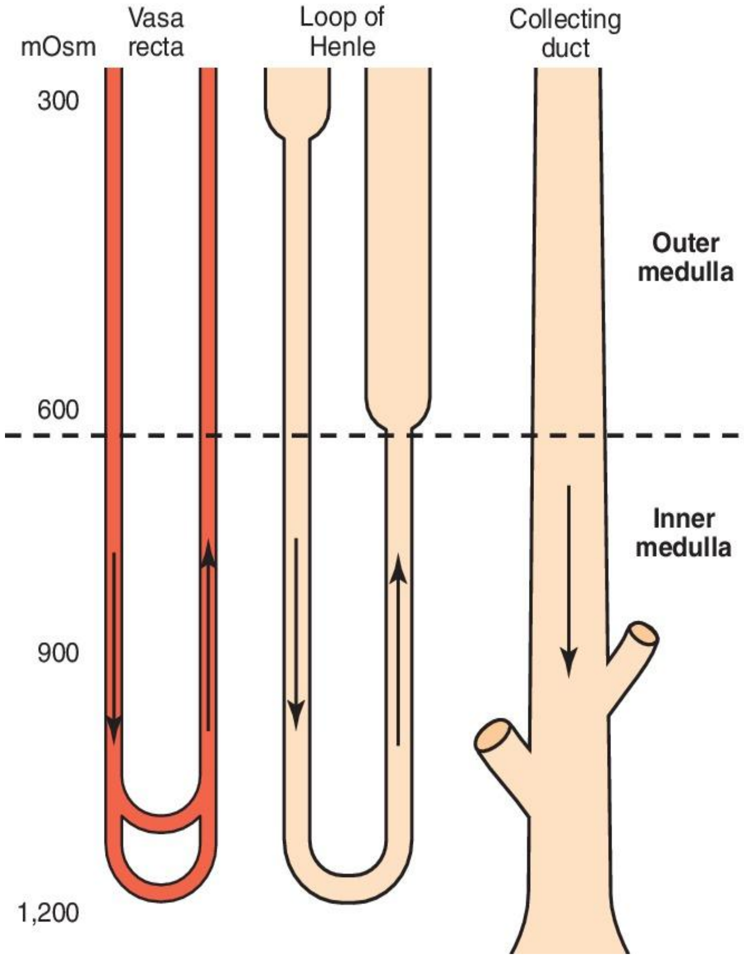

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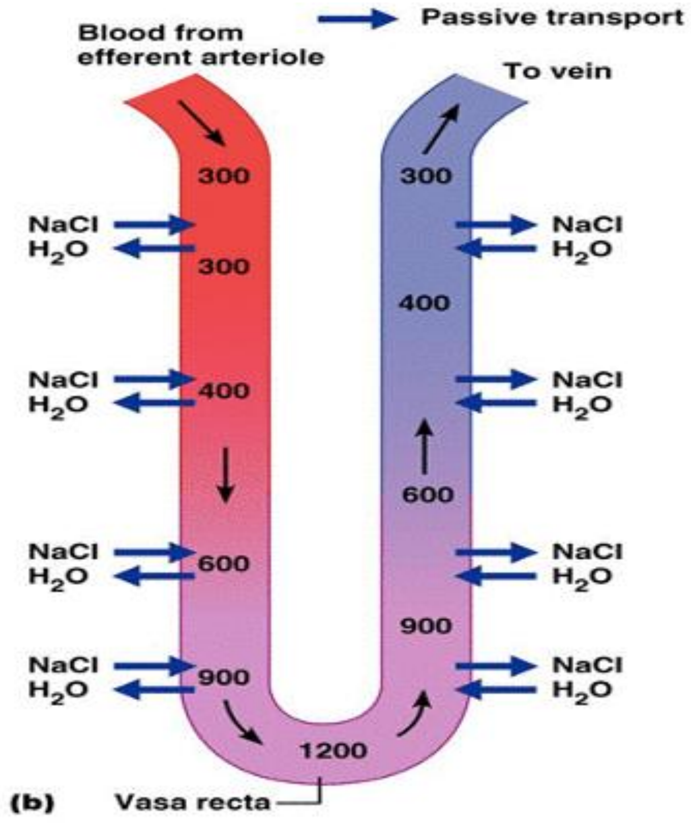
Countercurrent Exchanger – Vasa Recta

Downstream – NaCl diffuses into, and water diffuses out of the vasa recta increasing the osmolality.

Upstream – NaCl diffuses out, and water diffuses into the vasa recta, restoring the osmolality to 300 mOsm as blood enters the venous circulation.

The movement of NaCl and water is passive and does not require energy.



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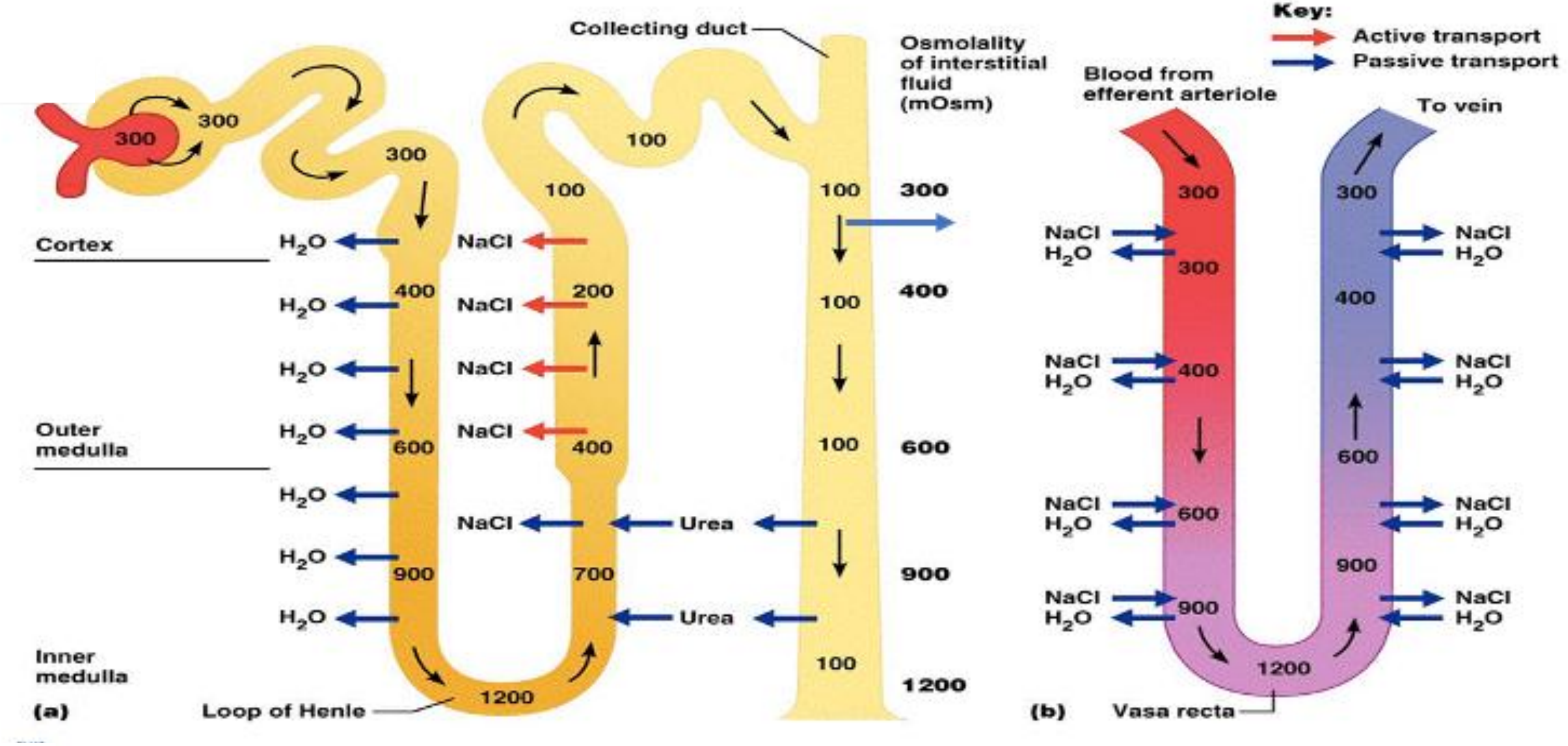
Countercurrent Multiplication and Exchanger



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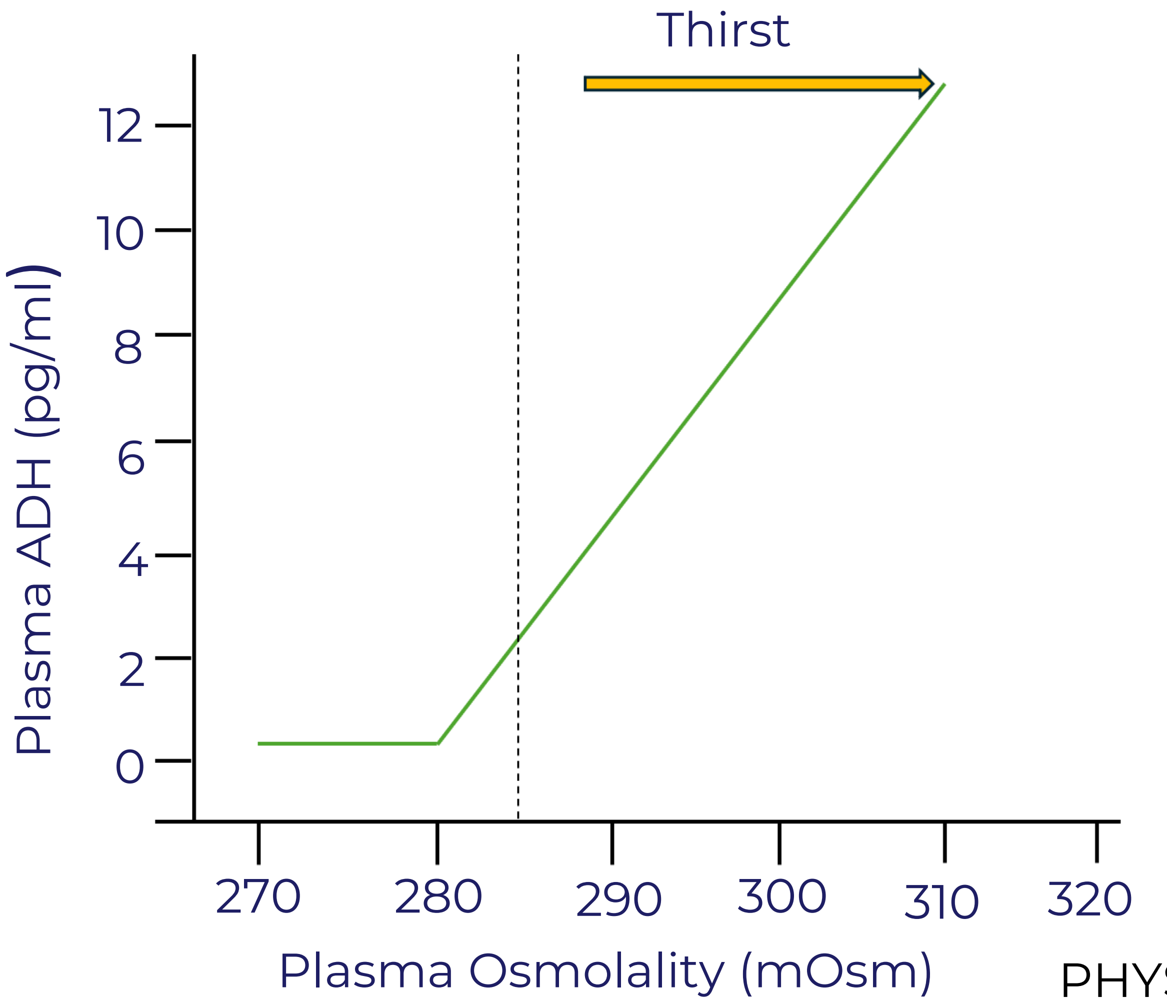
Anti-Diuretic Hormone - ADH

ADH (also called vasopressin) is produced in the hypothalamus and stored in the posterior pituitary.

An increase in plasma osmolality is the major mechanism responsible for increased ADH release.

Other factors that increase ADH release include significant fluid loss, trauma, emotional stress, etc.

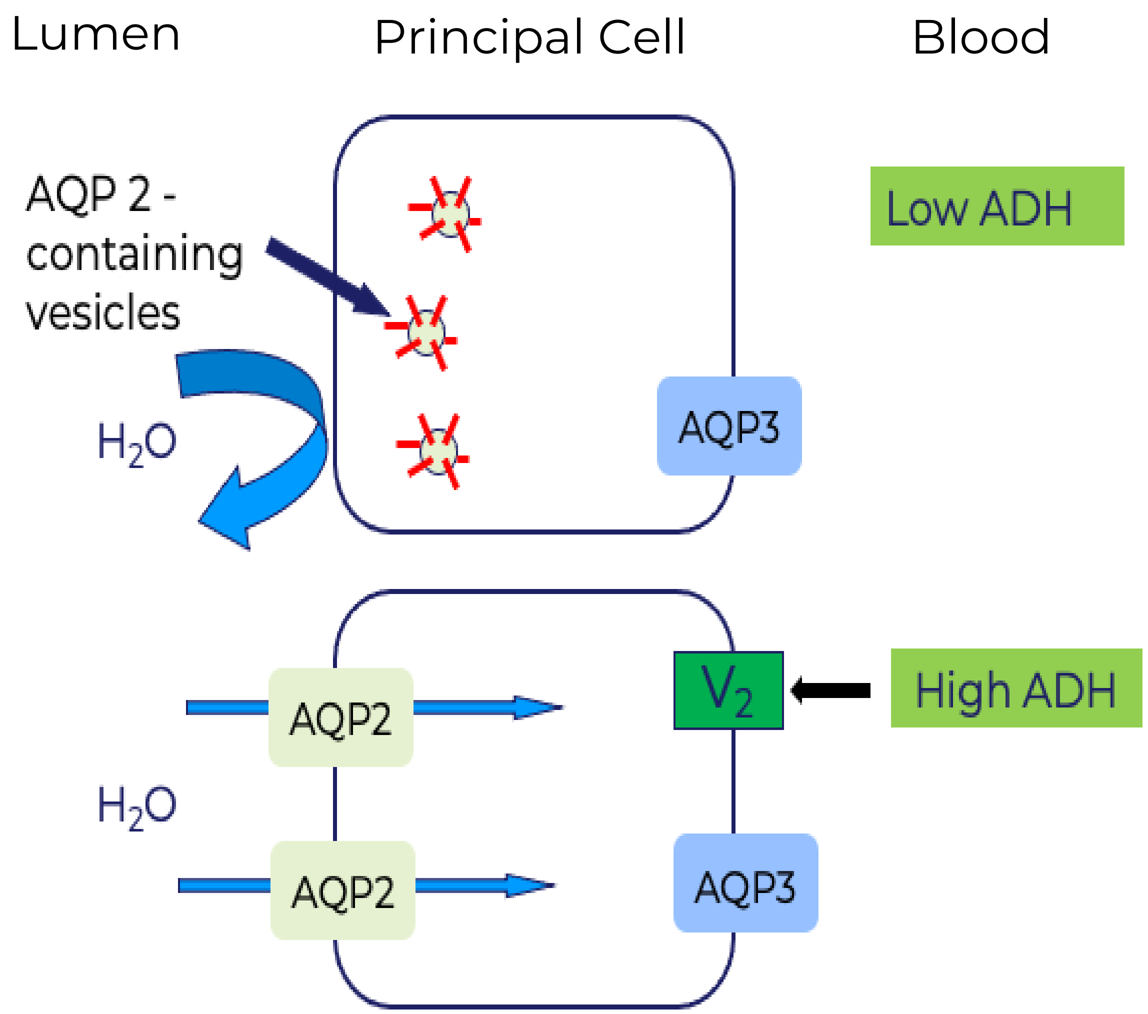


ADH and Water Reabsorption

ADH binds to V2 receptors on the principal cells of the collecting ducts.

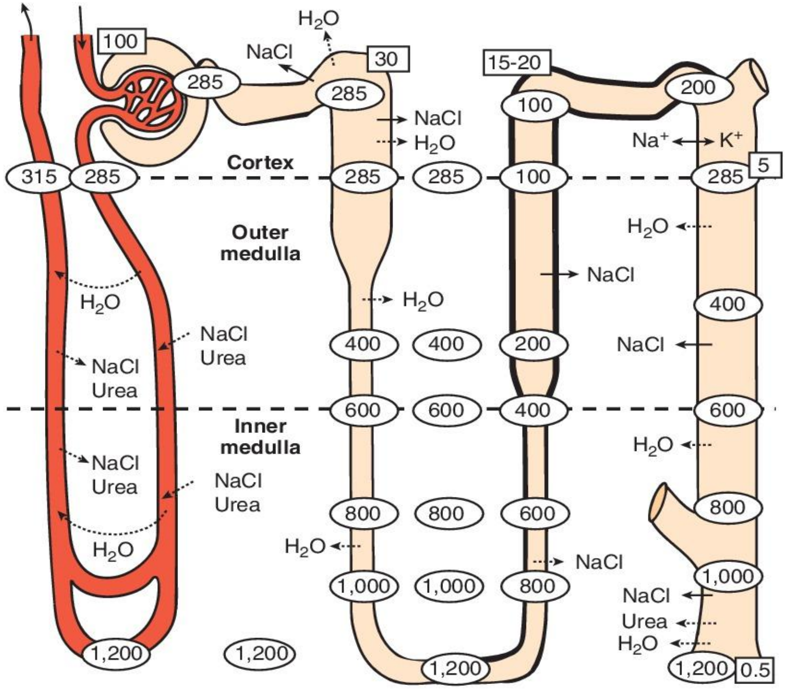
This results in a rapid insertion of AQP2 channels on the apical membrane of the principal cell, allowing for water reabsorption.

Water leaves the cell via AQP3 on the basolateral cell membrane.

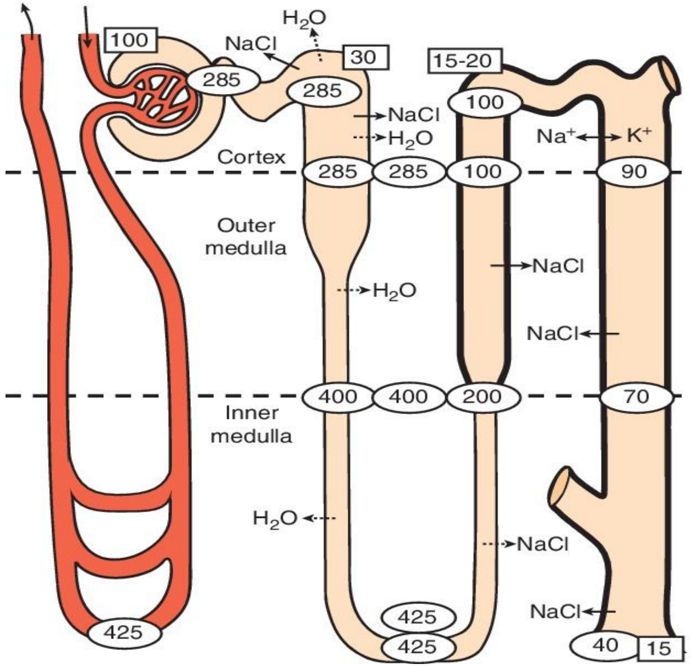


Concentrated Vs. Dilute Urine

Concentrated Urine
ADH is high.
The CDs are permeable to water.
Interstitial fluid is significantly concentrated.
Urea reabsorption in the inner MCD is increased.



Dilute Urine
ADH is low.
Water reabsorption is reduced.
The osmotic gradient in the medulla is reduced.
Urea reabsorption in the inner MCD is decreased.



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Diabetes Insipidus

Diabetes insipidus (DI) is due to a lack of ADH (central DI) or unresponsiveness of the kidneys to ADH (nephrogenic DI).

Water reabsorption fails, resulting in increased amounts of dilute urine and increased thirst.

Central DI
- Damage to the pituitary gland or hypothalamus (e.g., trauma)
- Inherited disorder

Nephrogenic DI
- Chronic kidney disease
- Medications (e.g., lithium)
- Inherited disorder

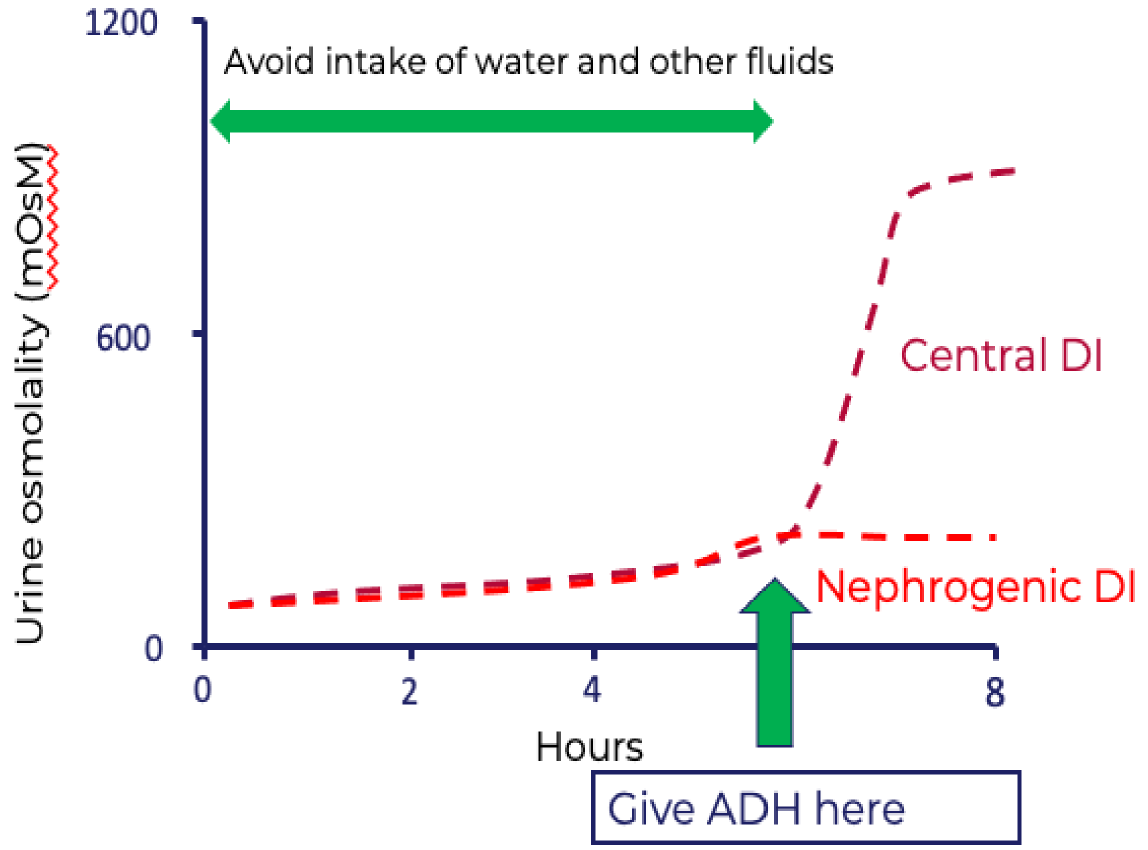
Central DI or Nephrogenic DI?

Water Deprivation Test
Water and other fluids are avoided for several hours. During this time, in DI, large amounts of dilute urine is still passed.

ADH is then administered.

In central DI, urine volume decreases and urine osmolality increases.

In nephrogenic DI, urine volume remains increased and urine osmolality remains decreased.



Syndrome of Inappropriate Antidiuretic Hormone Secretion (SIADH)

There is an **excessive** secretion of ADH.

Water reabsorption in the kidneys is increased significantly.

Patients with this condition tend to pass a concentrated urine (hyperosmotic urine).

Causes
- CNS infections
- Trauma
- Lung infections
- Cancers
- Drugs

References

- Medical Physiology: Principles for Clinical Medicine, 6e
- Ganong's Review of Medical Physiology, 26e